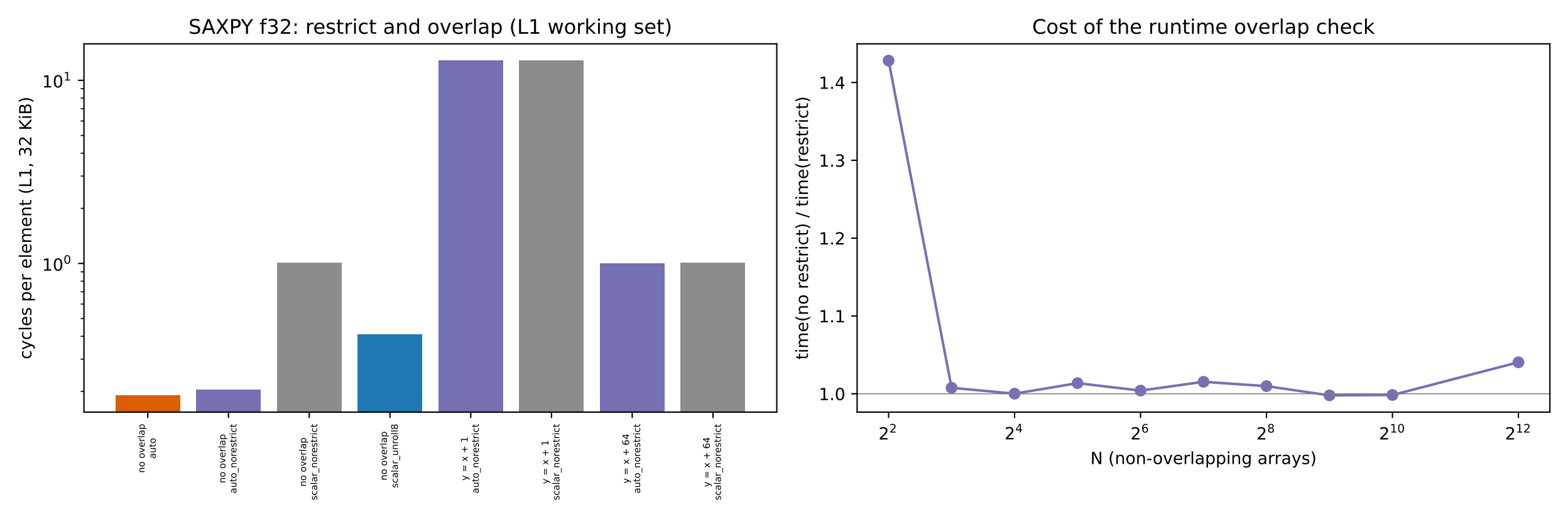
SAXPY f32: restrict and overlap (L1 working set)
cycles per element (L1, 32 KiB)
10^1
10^0
no overlap auto
no overlap auto_norestrict
no overlap scalar_norestrict
no overlap scalar_unroll8
y = x + 1 auto_norestrict
y = x + 1 scalar_norestrict
y = x + 64 auto_norestrict
y = x + 64 scalar_norestrict
Cost of the runtime overlap check
time(no restrict) / time(restrict)
1.4
1.3
1.2
1.1
1.0
2^2
2^4
2^6
2^8
2^{10}
2^{12}
N (non-overlapping arrays)